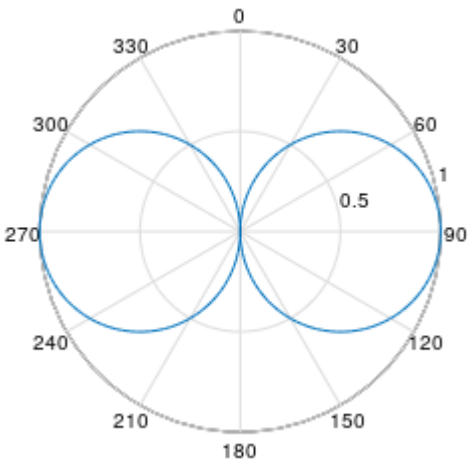
TUGAS BESAR ANTENA & PROPAGASI

Nama: Axel Ega Adiputra (201344003)

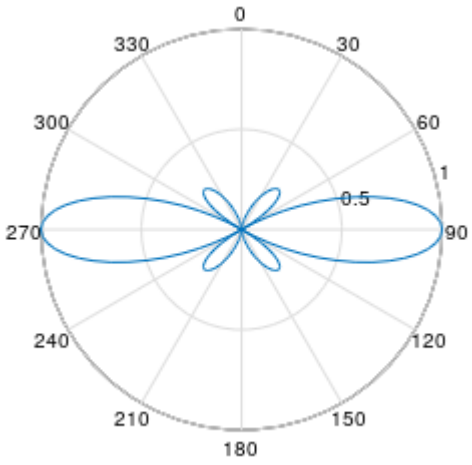
Bobby Widiya Dwi Elputra (201344005)

Kelas: 3NK

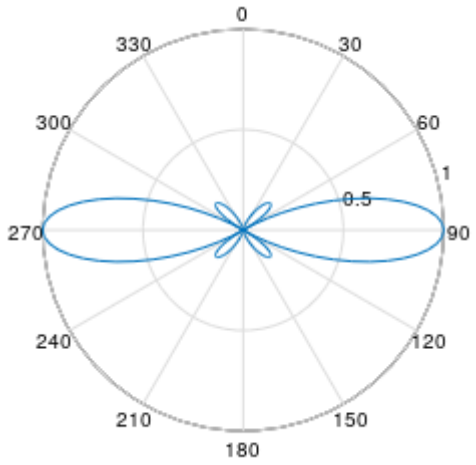
1. Sebuah antena susun (array) broadside terdiri dari 4 buah elemen antena dipol ideal. Jika jarak antar elemen $d=\lambda 2/$, tentukan dan plot pola elemen pola elemen $g(\theta,\phi)$, pola faktor susun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ untuk masing-masing beda fasa antar elemen α:
   a) $\alpha = 0°$
      - Pola elemen g(**θ,φ**)



      - Pola factor susun f(**θ,φ**)

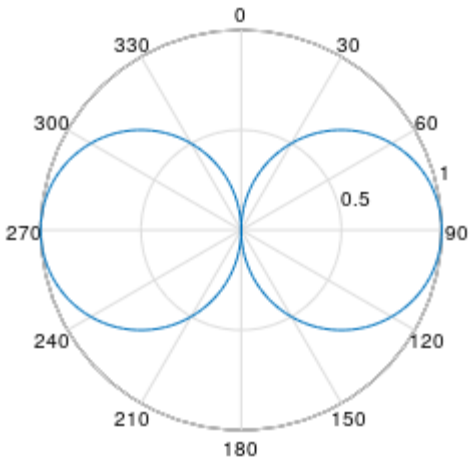


      - Pola total f(**θ,φ**)

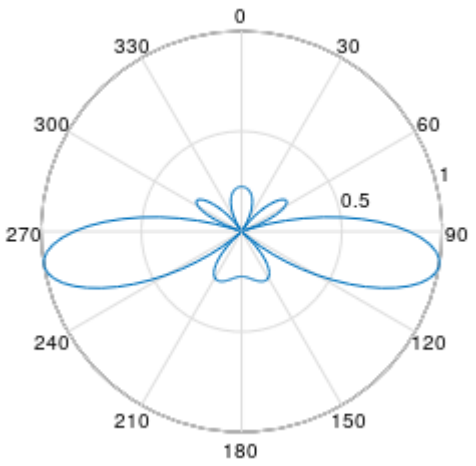


b) a = 30°

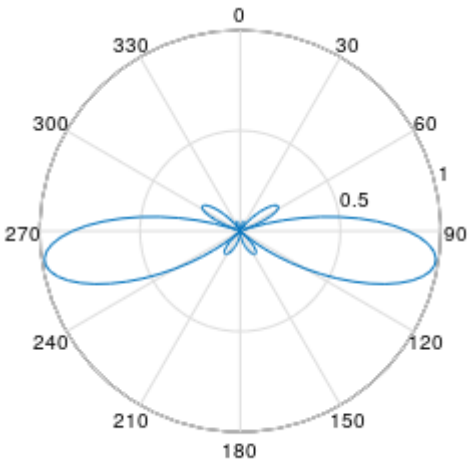
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

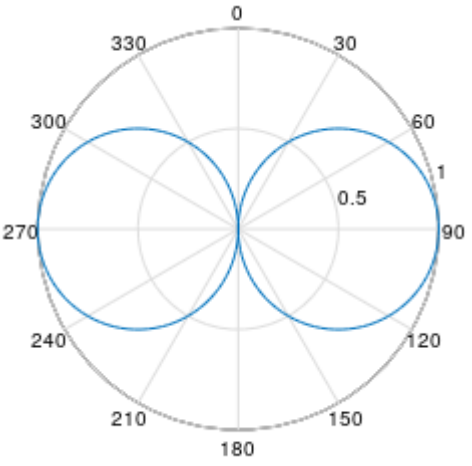


- Pola total f(**θ,φ**)

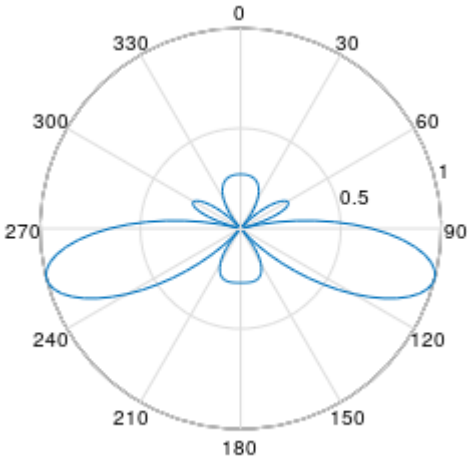


c) a = 45°

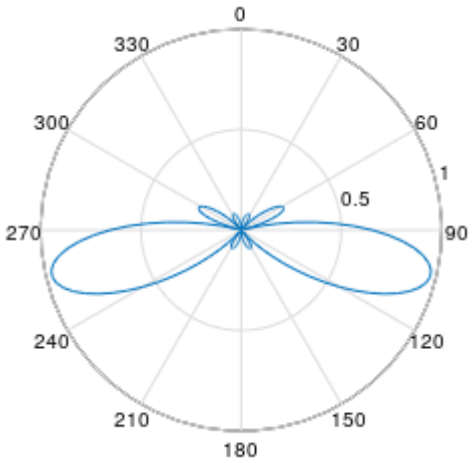
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

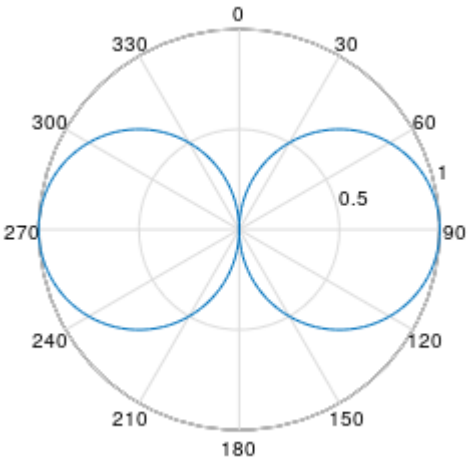


- Pola total f(**θ,φ**)

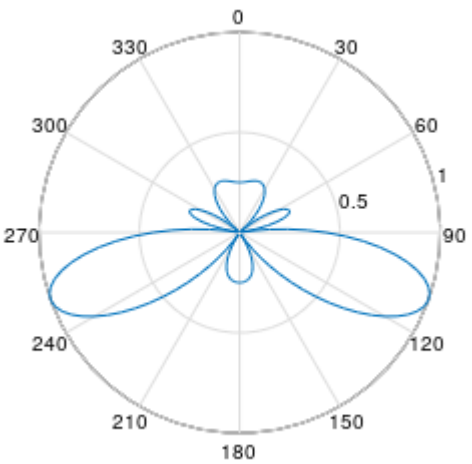


d) a = 60°

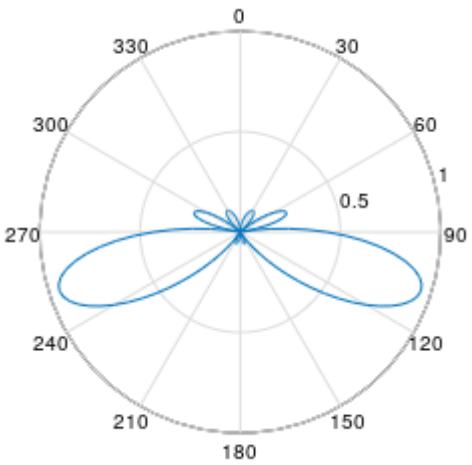
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

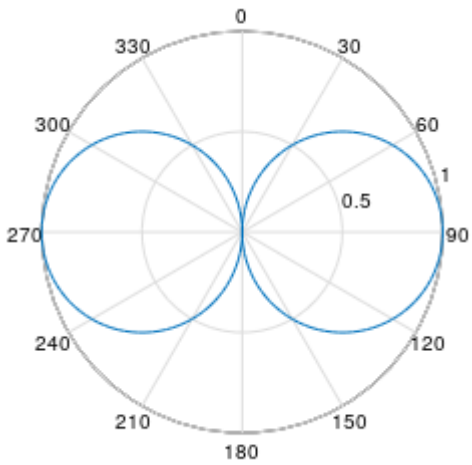


- Pola total f(**θ,φ**)

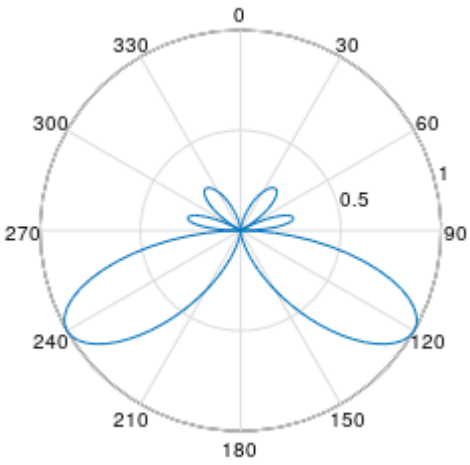


e) a = 90°

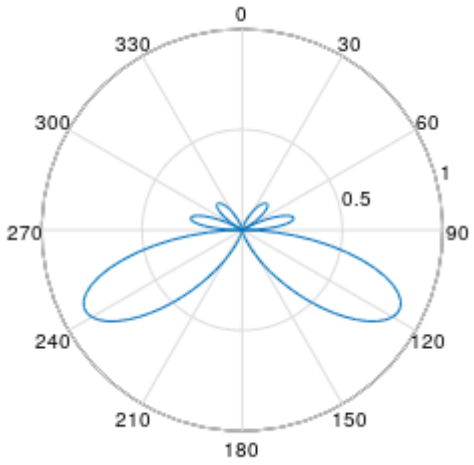
- Pola elemen g(**θ,φ**)



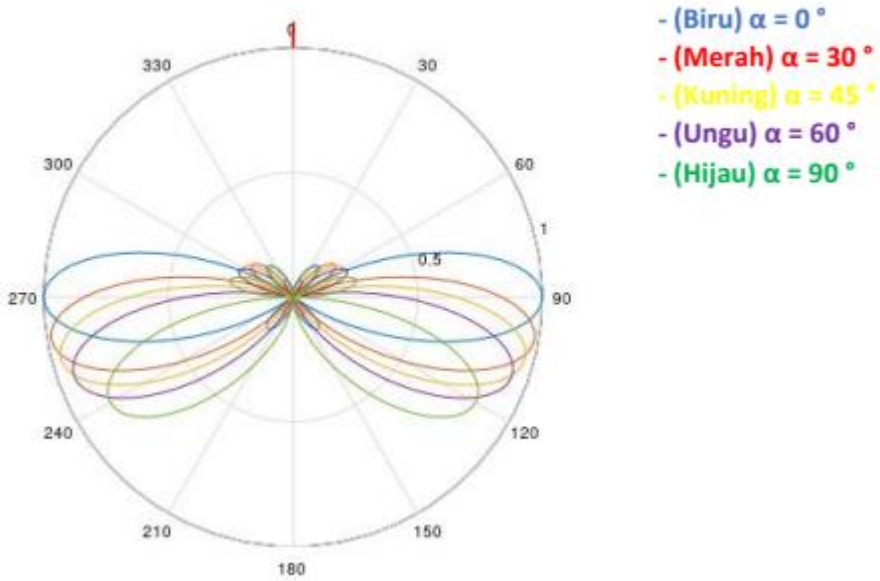
- Pola factor susun f(**θ,φ**)



- Pola total f(**θ,φ**)



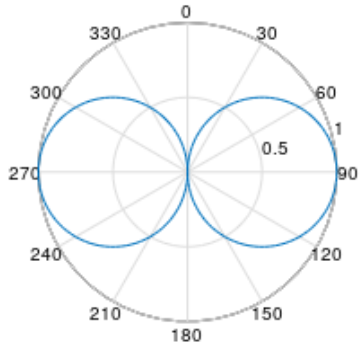
Gabungan Plot Total F(θ,φ)



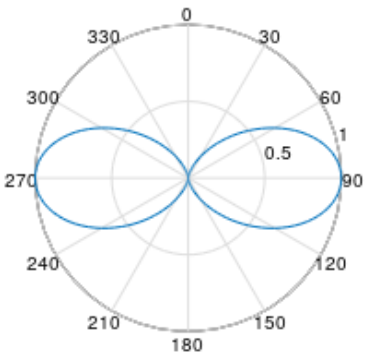
2. Sebuah antena susun (array) broadside terdiri dari 4 buah elemen antena dipol ideal. Jika beda fasa antar elemen α=0°, tentukan dan plot pola elemen pola elemen $g(\theta,\phi)$, pola faktor susun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ masing-masing untuk jarak antar elemen d:

   a) $d = \lambda/4$

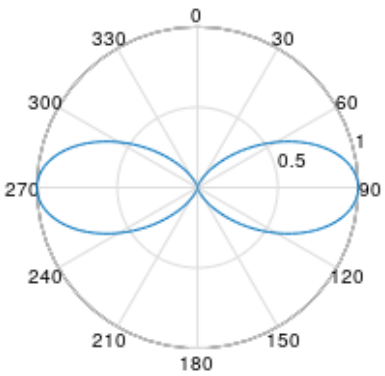
   - Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

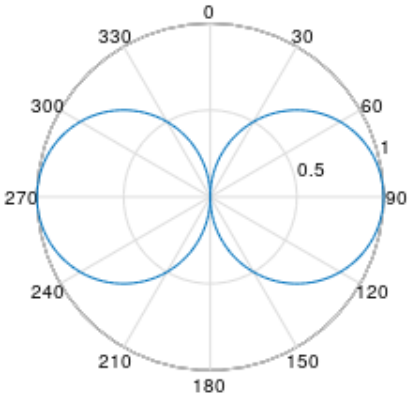


- Pola total f(**θ,φ**)

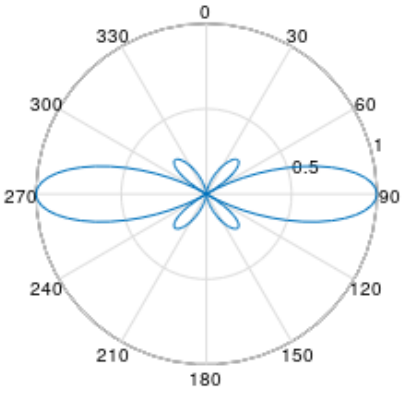


b) d = λ/2

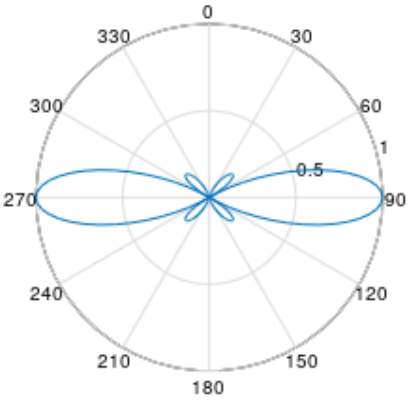
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

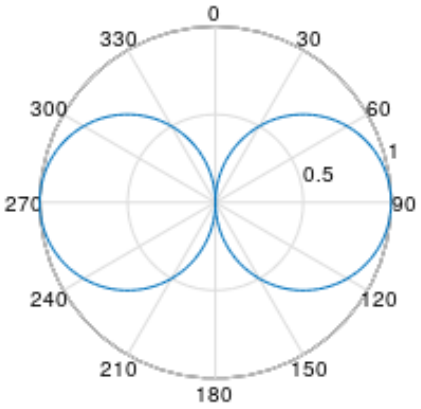


- Pola total f(**θ,φ**)

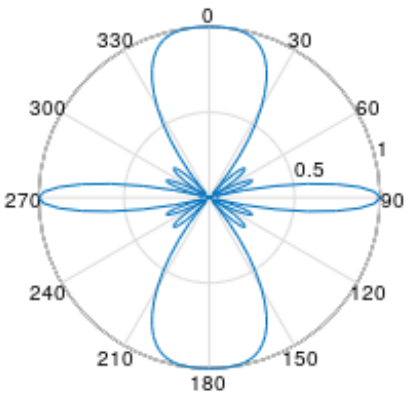


c) $d = \lambda$

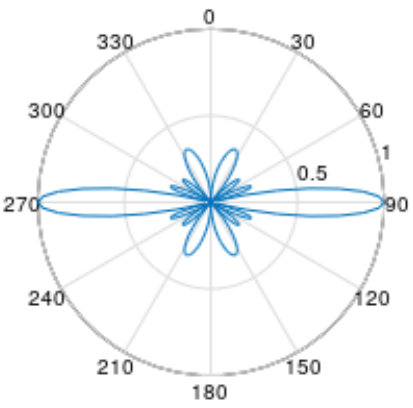
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

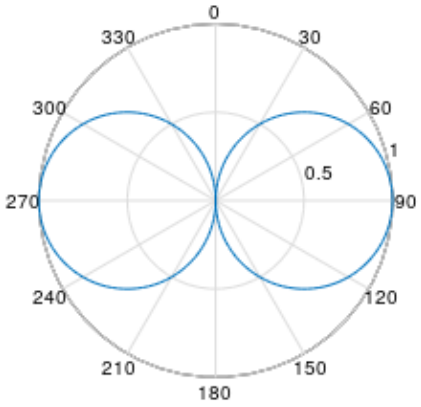


- Pola total f(**θ,φ**)

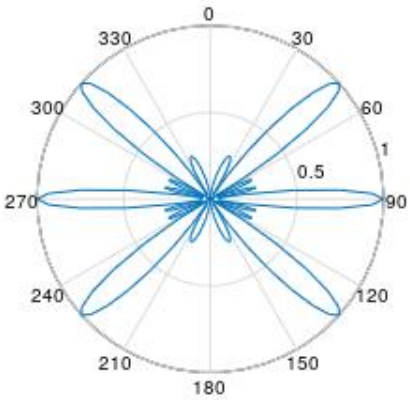


d) d = 3λ/2

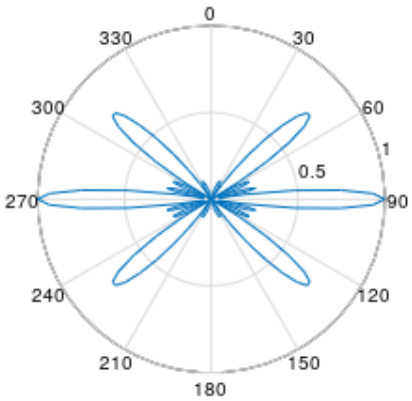
- Pola elemen g(**θ,φ**)



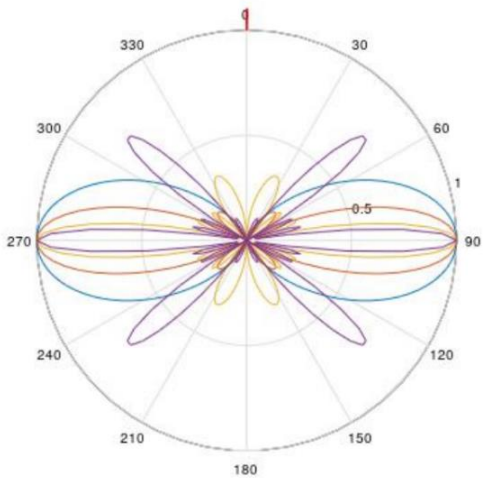
- Pola factor susun f(**θ,φ**)



- Pola total f(**θ,φ**)



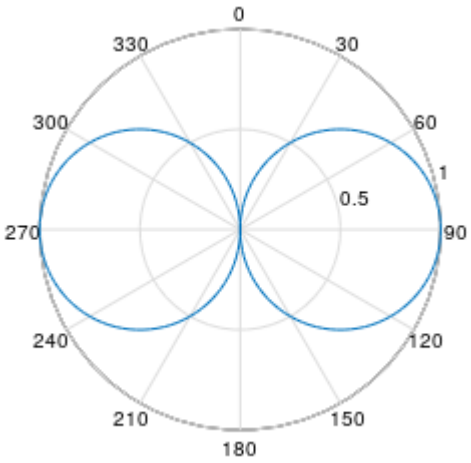
Gabungan Plot Total F(θ,φ)



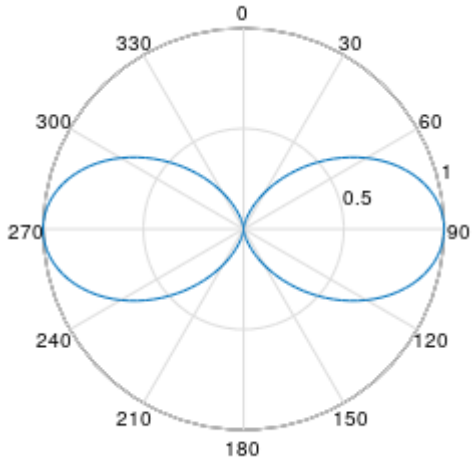
3. Sebuah antena susun (array) broadside terdiri dari n buah elemen antena dipol ideal. Jika beda fasa antar elemen α=0°, dan jarak anter elemen $d=\lambda 2\diagup$, tentukan dan plot pola elemen pola elemen $g(\theta,\phi)$, pola faktor susun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ masing-masing untuk jumlah elemen n:

   a) *n = 2*

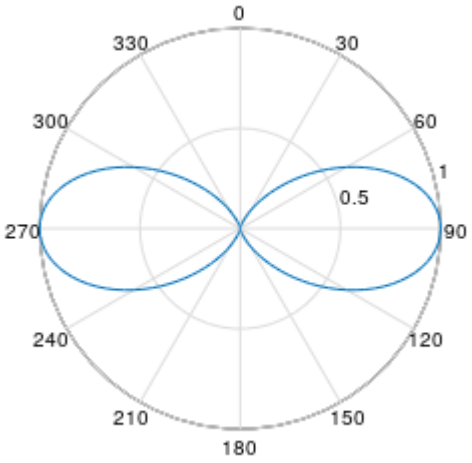
   - Pola elemen g(**θ,φ**)



   - Pola factor susun f(**θ,φ**)

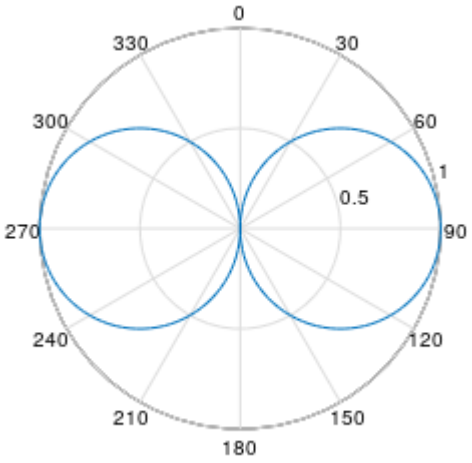


- Pola total f(**θ,φ**)

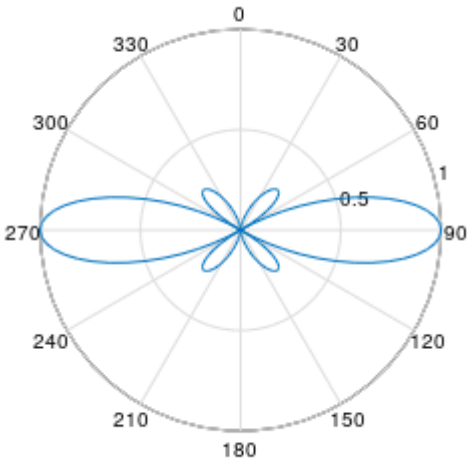


b) n = 4

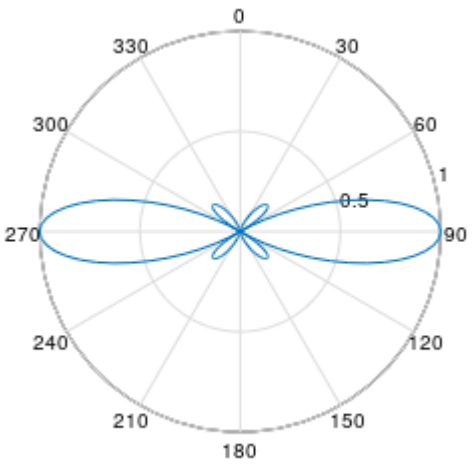
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

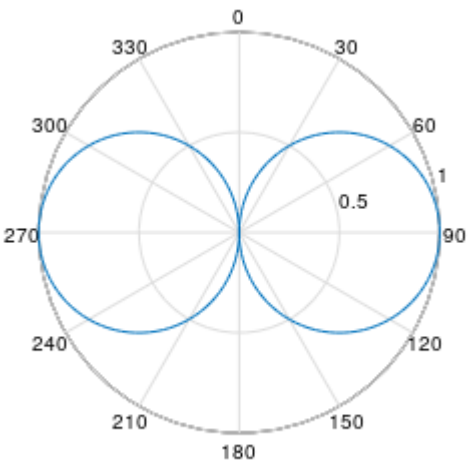


- Pola total f(**θ,φ**)

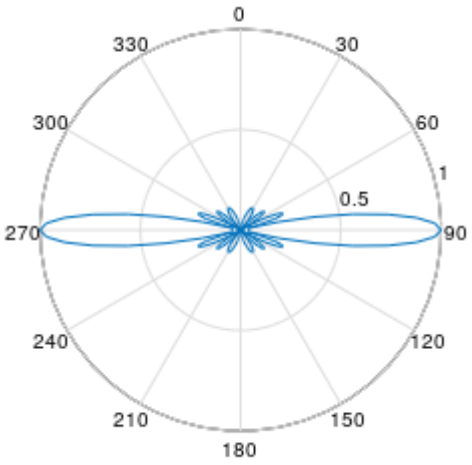


c) n = 8

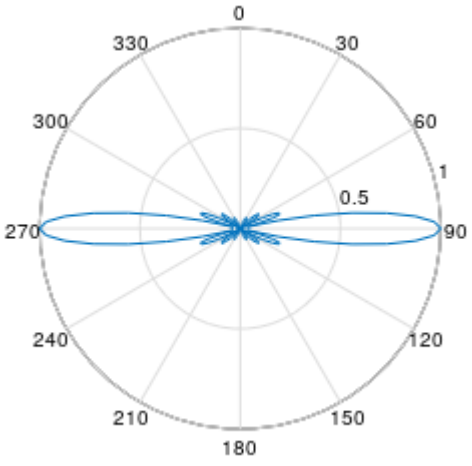
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

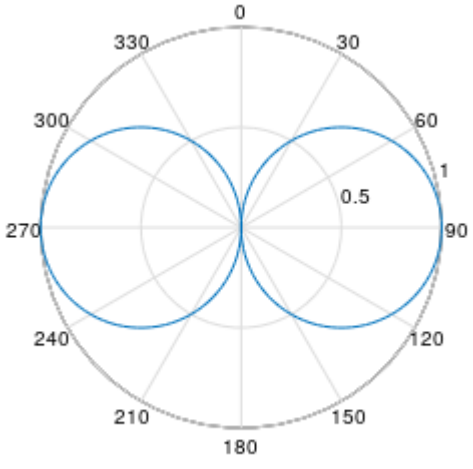


- Pola total f(**θ,φ**)

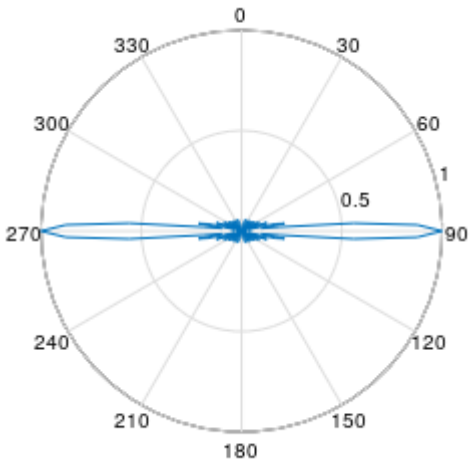


d) n = 16

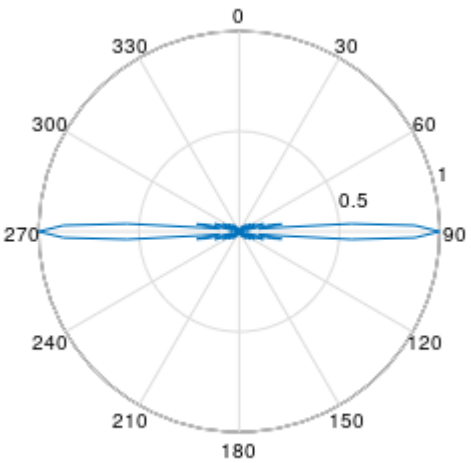
- Pola elemen g(**θ,φ**)



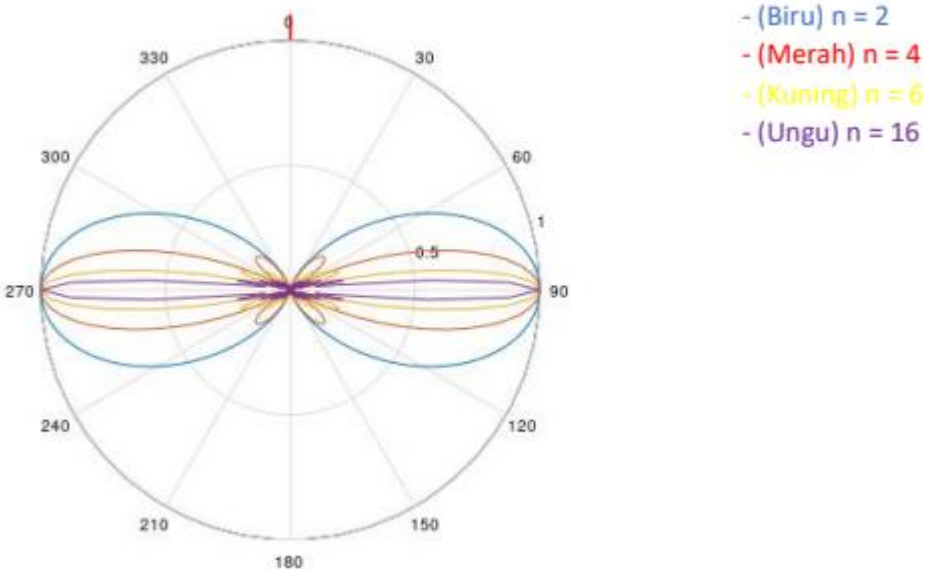
- Pola factor susun f(**θ,φ**)



- Pola total f(**θ,φ**)



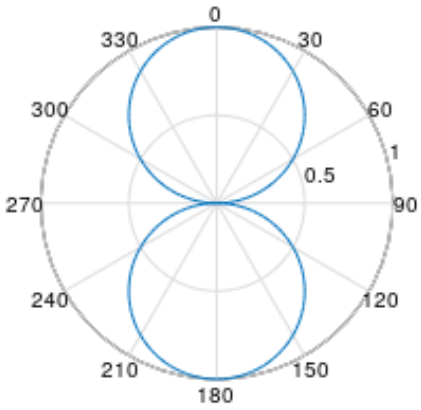
Gabungan Plot Total F(θ,φ)



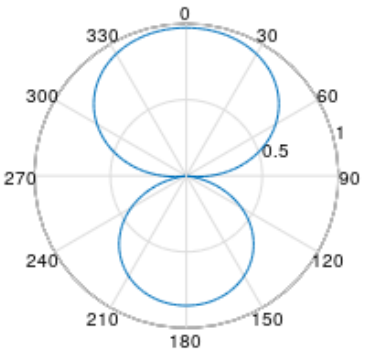
4. Sebuah antena susun (array) endfire terdiri dari n buah elemen antena dipol ideal. Antena susun ini diarahkan ke θ=0°. (Tetapkan pola elemennya adalah (θ,)=cosθ). Plot pola faktor susun $f(\theta,\phi)$ dan pola total $F(\theta,\phi)$ masing-masing untuk jumlah elemen n:

   a. n = 2

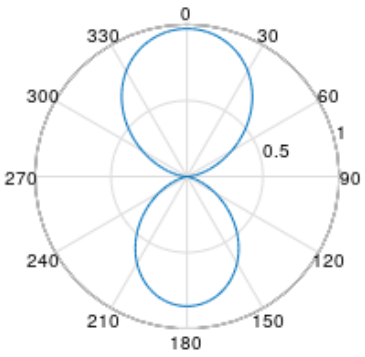
   - Pola elemen g(**θ,φ**)



   - Pola factor susun f(**θ,φ**)

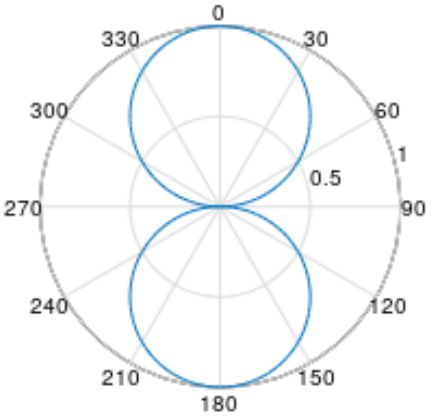


   - Pola total f(**θ,φ**)

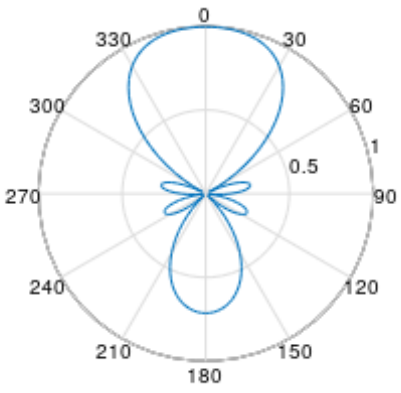


   b) n = 4

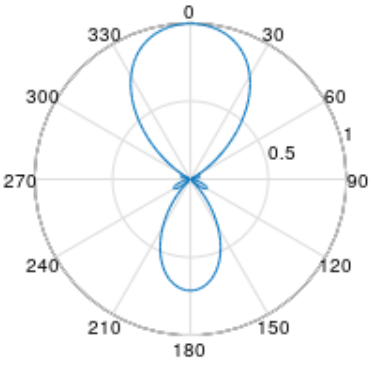
   - Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

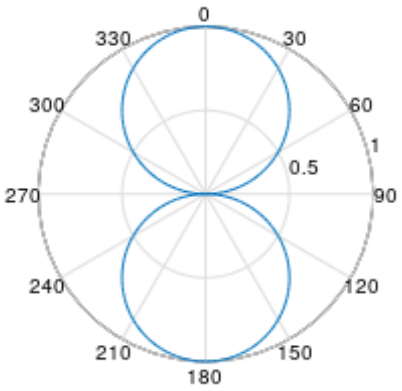


- Pola total f(**θ,φ**)

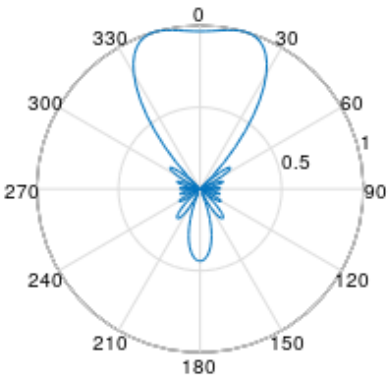


c) n = 8

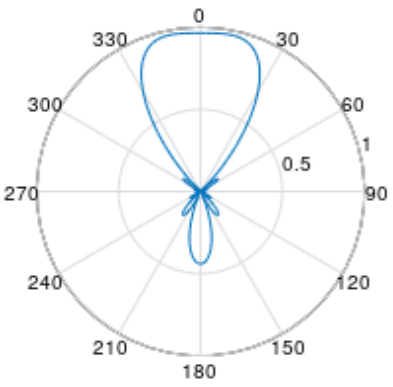
- Pola elemen g(**θ,φ**)



- Pola factor susun f(**θ,φ**)

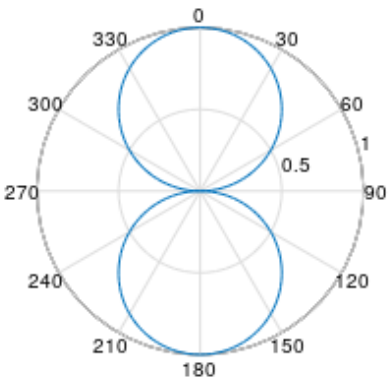


- Pola total f(**θ,φ**)

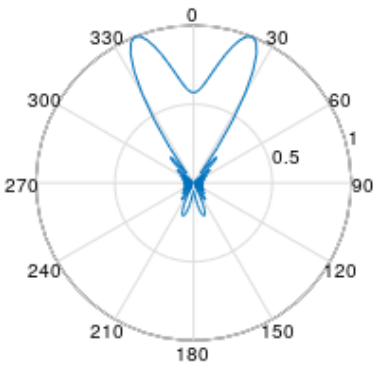


d) n = 16

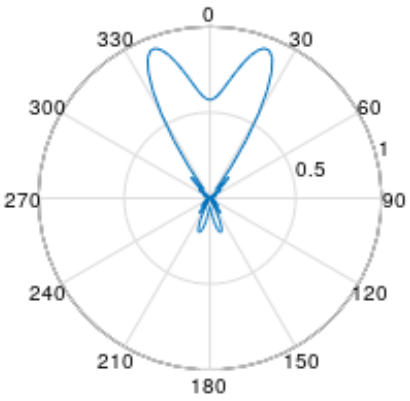
- Pola elemen g(**θ,φ**)



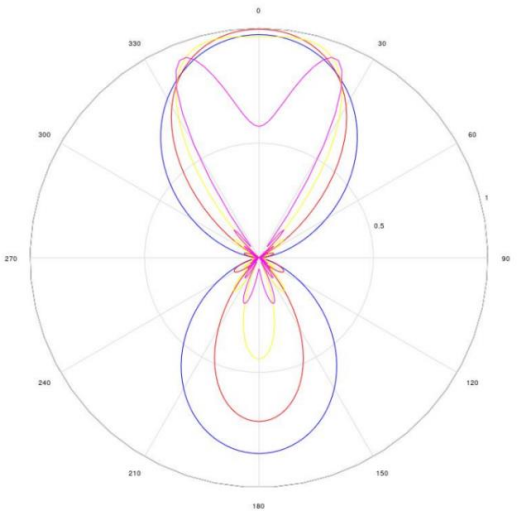
- Pola factor susun f(**θ,φ**)



- Pola total f(**θ,φ**)



Gabungan Pola total F(θ,φ)



- (Biru) n = 2
- (Merah) n = 4
- (Kuning) n = 6
- (Ungu) n = 16